

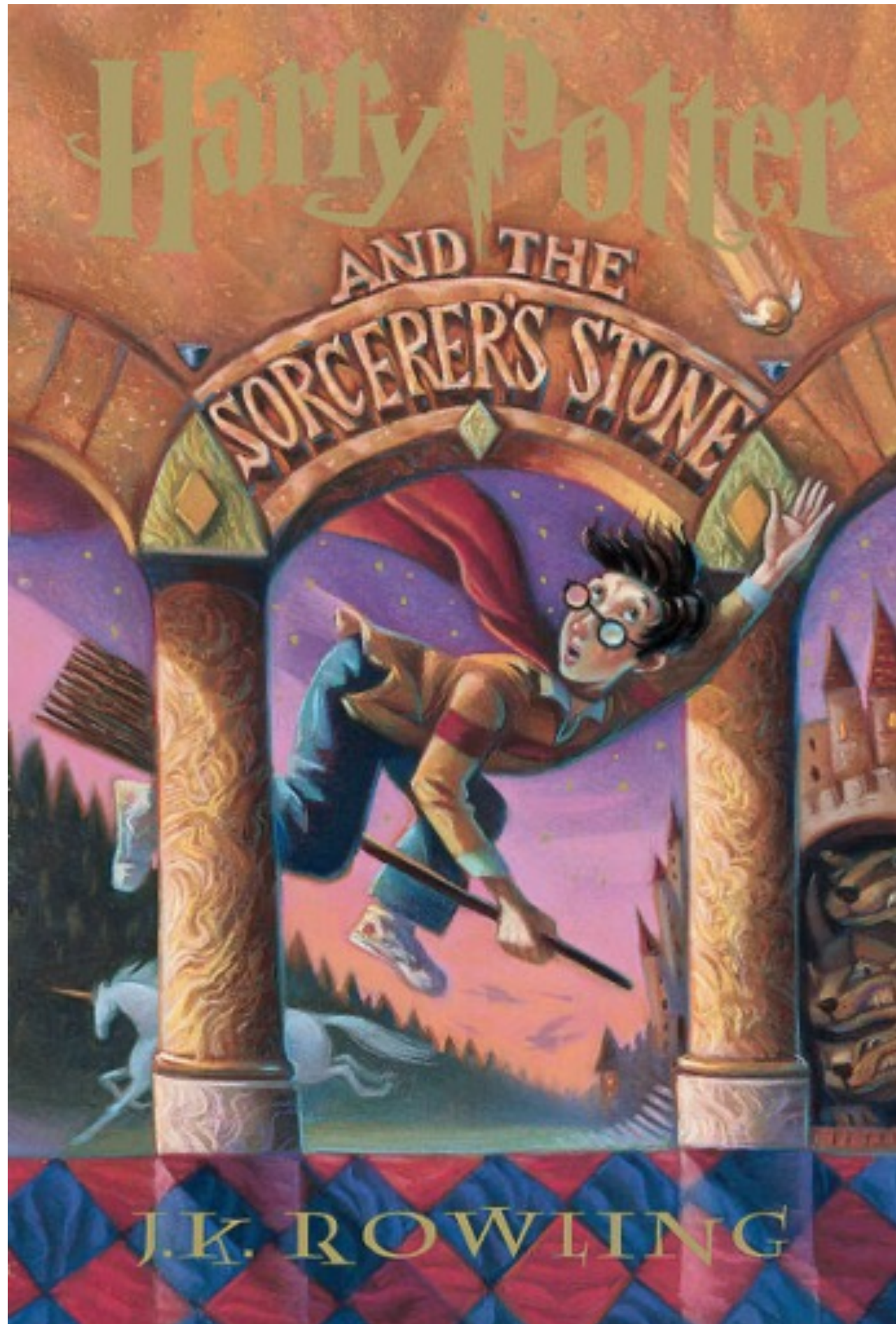
What Does It Mean for a Language Model to Forget?

Youssef Allouah

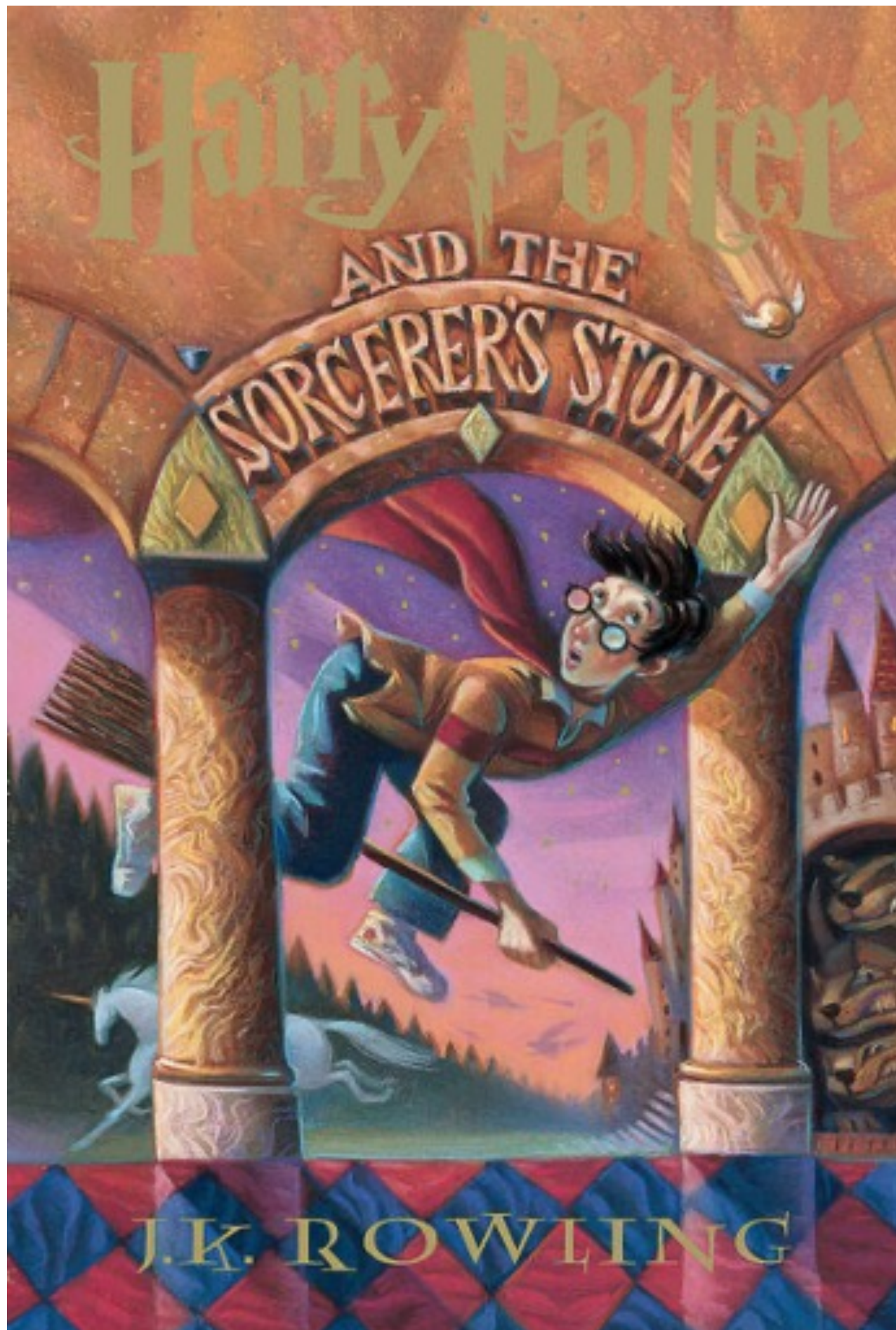
AdvML workshop, COLM 2026



“Forget Harry Potter”

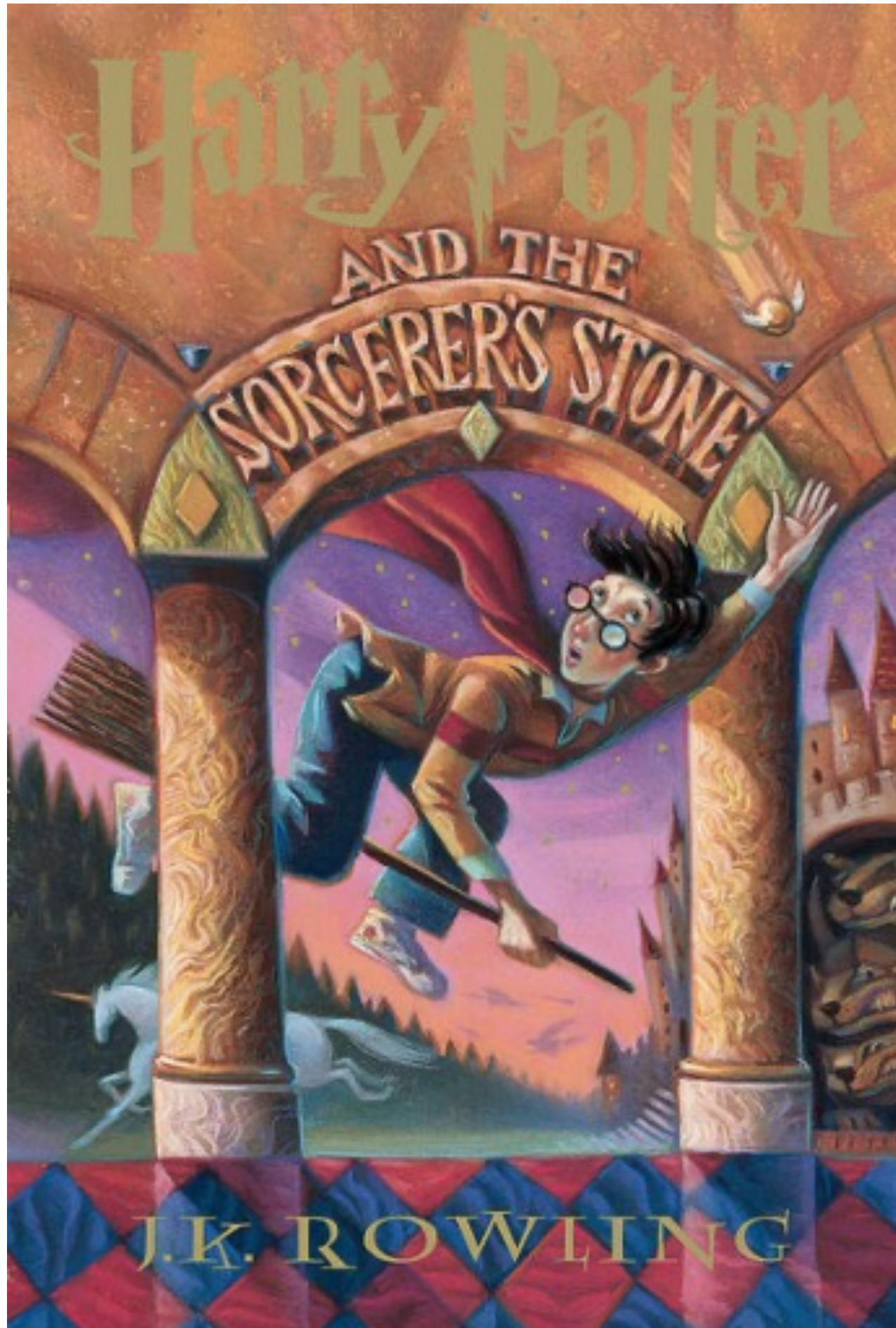


“Forget Harry Potter”



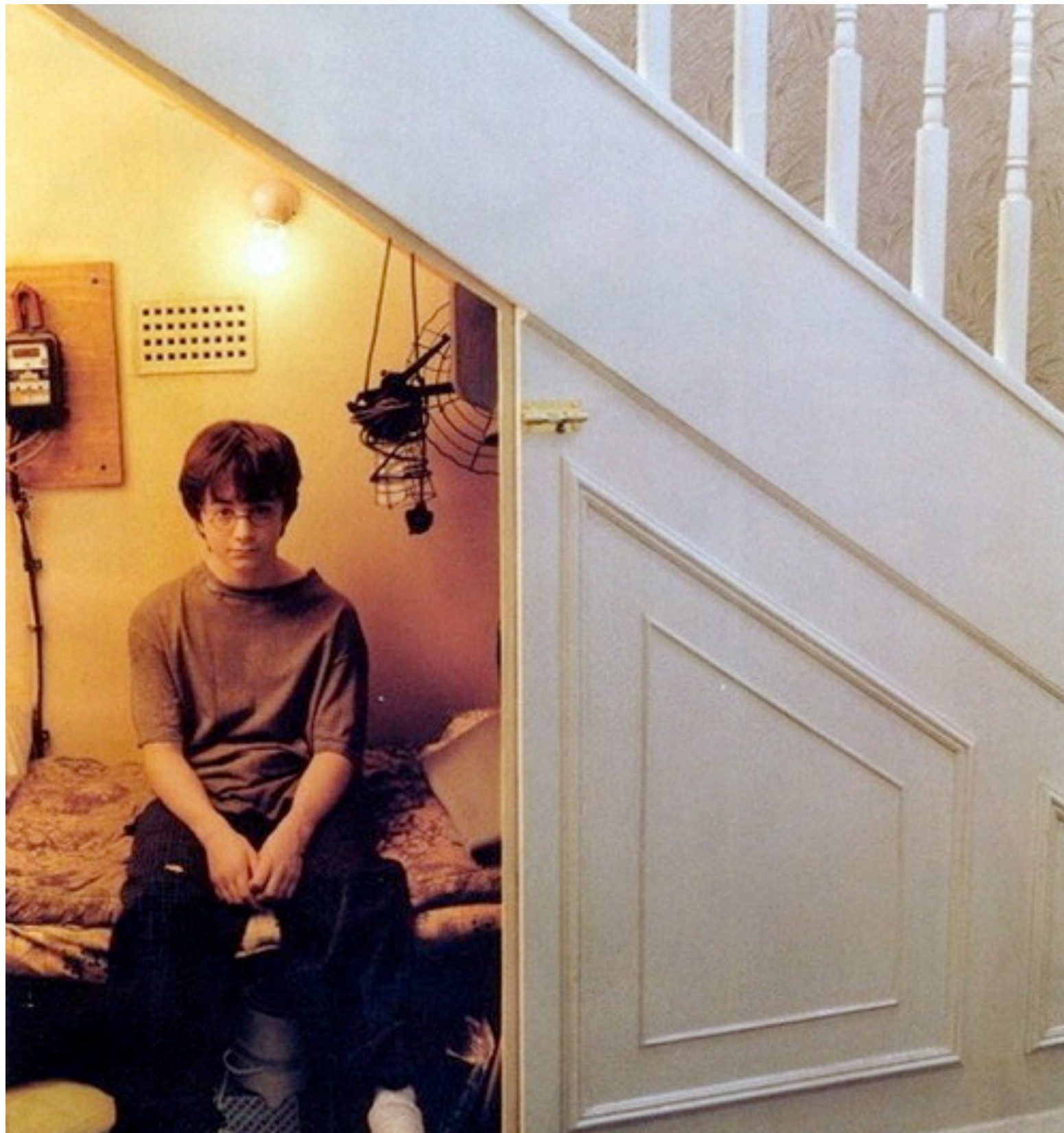
- Which of these abilities should the model lose?

“Forget Harry Potter”



- Which of these abilities should the model lose?
 1. **Characters:** “Where do the Dursleys make Harry sleep?”

“Forget Harry Potter”

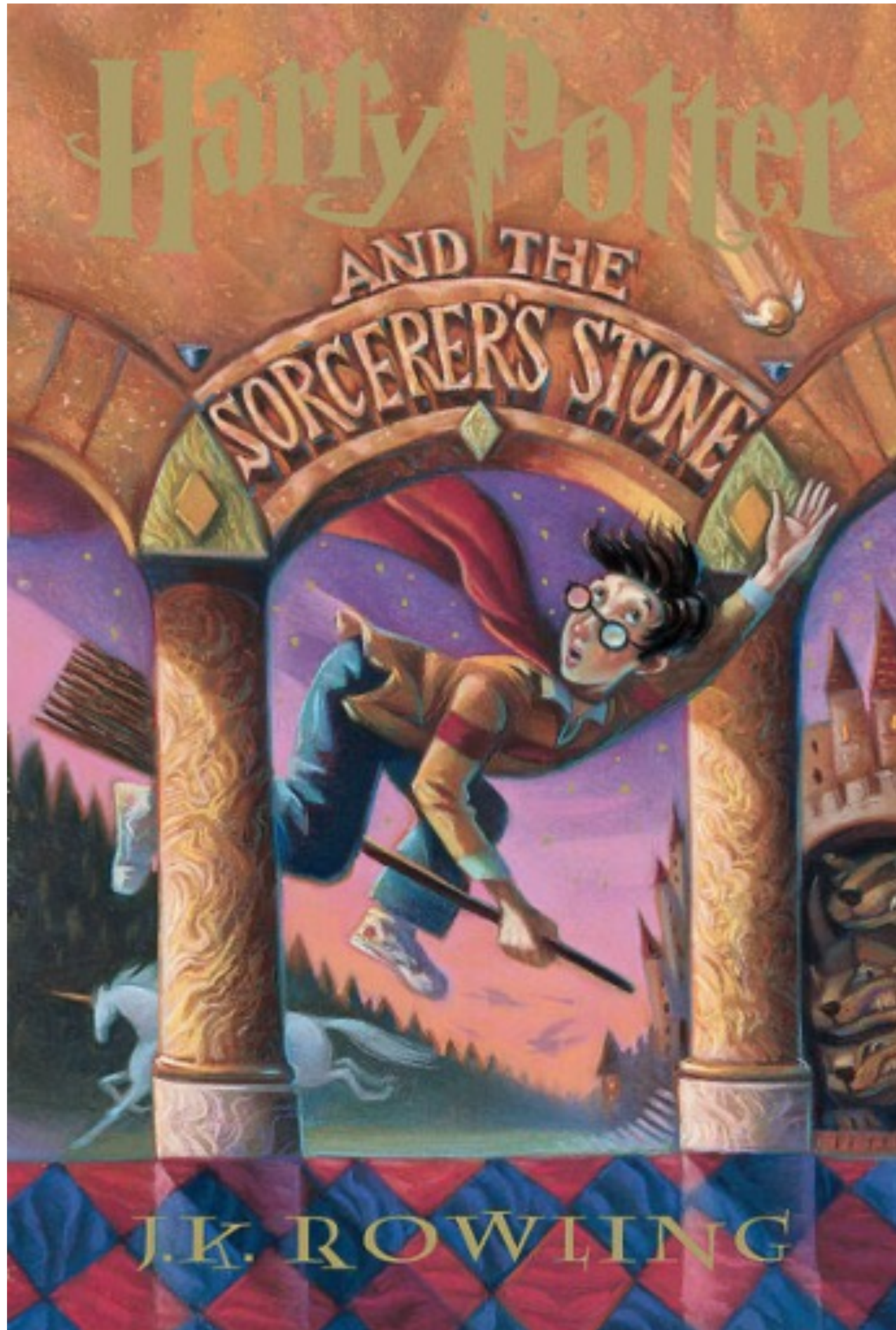


- Which of these abilities should the model lose?

1. **Characters:** “Where do the Dursleys make Harry sleep?”

Answer: Cupboard under the stairs.

“Forget Harry Potter”



- Which of these abilities should the model lose?
 1. Characters: “Where do the Dursleys make Harry sleep?”
 2. **Style:** “Write a new scene at a magical boarding school.”

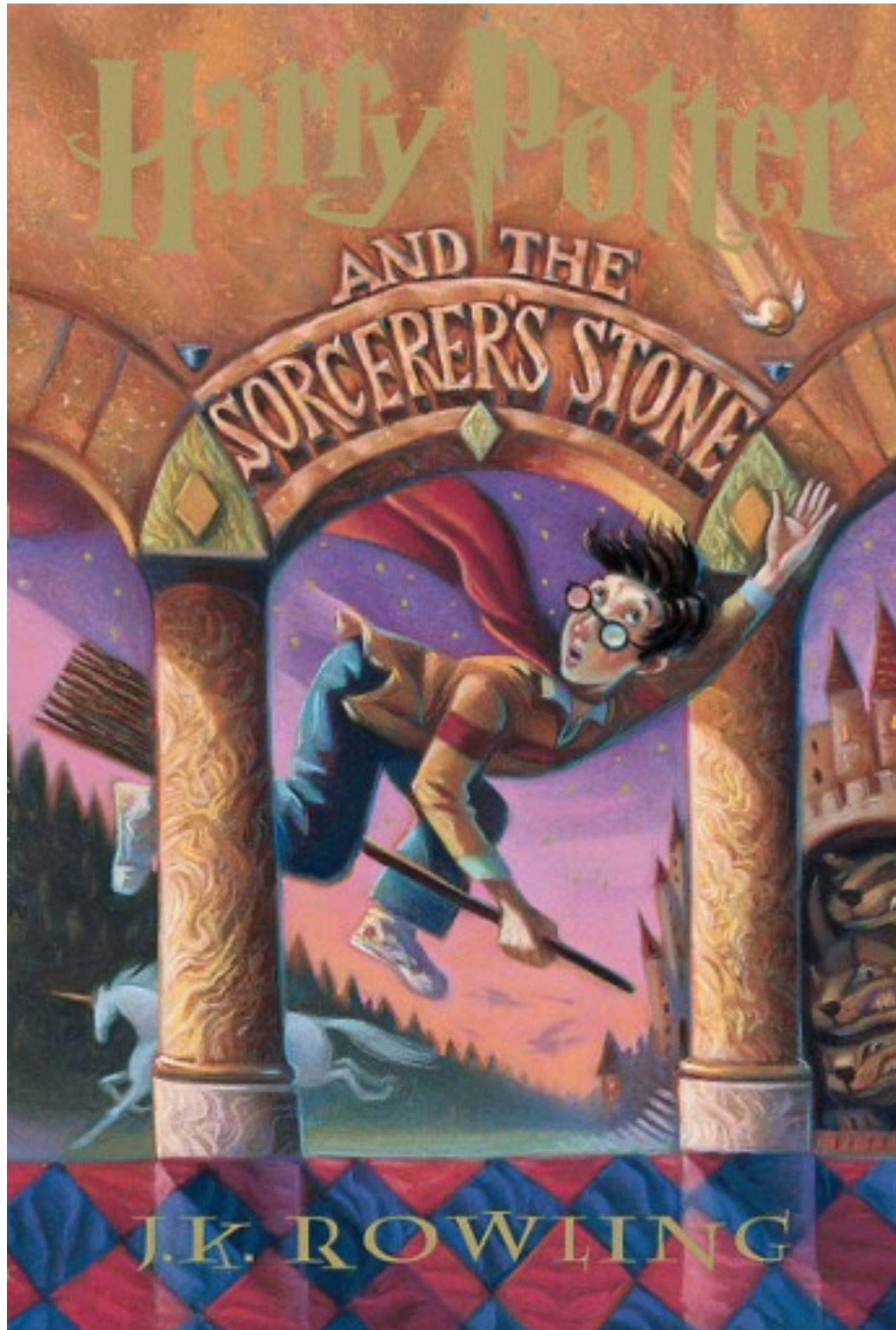
“Forget Harry Potter”



- Which of these abilities should the model lose?
 1. Characters: “Where do the Dursleys make Harry sleep?”
 2. **Style:** “Write a new scene at a magical boarding school.”

Potential answer: “The staircase changed its mind halfway to the classroom.”

“Forget Harry Potter”



- Which of these abilities should the model lose?
 1. Characters: “Where do the Dursleys make Harry sleep?”
 2. Style: “Write a new scene at a magical boarding school.”
 3. **Passages:** “Continue a passage from chapter 1.”

“Delete These Records”

- Data deletion has a clearer objective: approximate retraining (fast)

“Delete These Records”

- Data deletion has a clearer objective: approximate retraining (fast)

$$\text{Unlearn}(\text{Train}(\mathcal{D}_{\text{full}}); \mathcal{D}_{\text{remove}}) \approx_{\varepsilon} \text{Train}(\mathcal{D}_{\text{full}} \setminus \mathcal{D}_{\text{remove}})$$

“Delete These Records”

- We understand data deletion pretty well for canonical tasks (regression)

**Remember What You Want to Forget:
Algorithms for Machine Unlearning**

Near-Optimal Machine Unlearning Utility for Smooth Strongly
Convex Losses

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Published as a conference paper at ICLR 2025

THE UTILITY AND COMPLEXITY OF IN- AND OUT-OF-
DISTRIBUTION MACHINE UNLEARNING

arXiv:2412.09119

Youssef Allouah^{1*}, Joshua Kazdan², Rachid Guerraoui¹, Sanmi Koyejo²

¹EPFL, Switzerland ²Stanford University, USA

“Delete These Records”

- We are still trying to figure it out for more modern tasks

In 42nd *IEEE Symposium of Security and Privacy*

Machine Unlearning

Lucas Bourtoutle^{*‡§}, Varun Chandrasekaran^{*†}, Christopher A. Choquette-Choo^{*‡§}, Hengrui Jia^{*‡§},
Adelin Travers^{*‡§}, Baiwu Zhang^{*‡§}, David Lie[‡], Nicolas Papernot^{‡§}
University of Toronto[‡], Vector Institute[§], University of Wisconsin-Madison[†]

Langevin Unlearning: A New Perspective of Noisy Gradient Descent for Machine Unlearning

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Certified Unlearning for Neural Networks

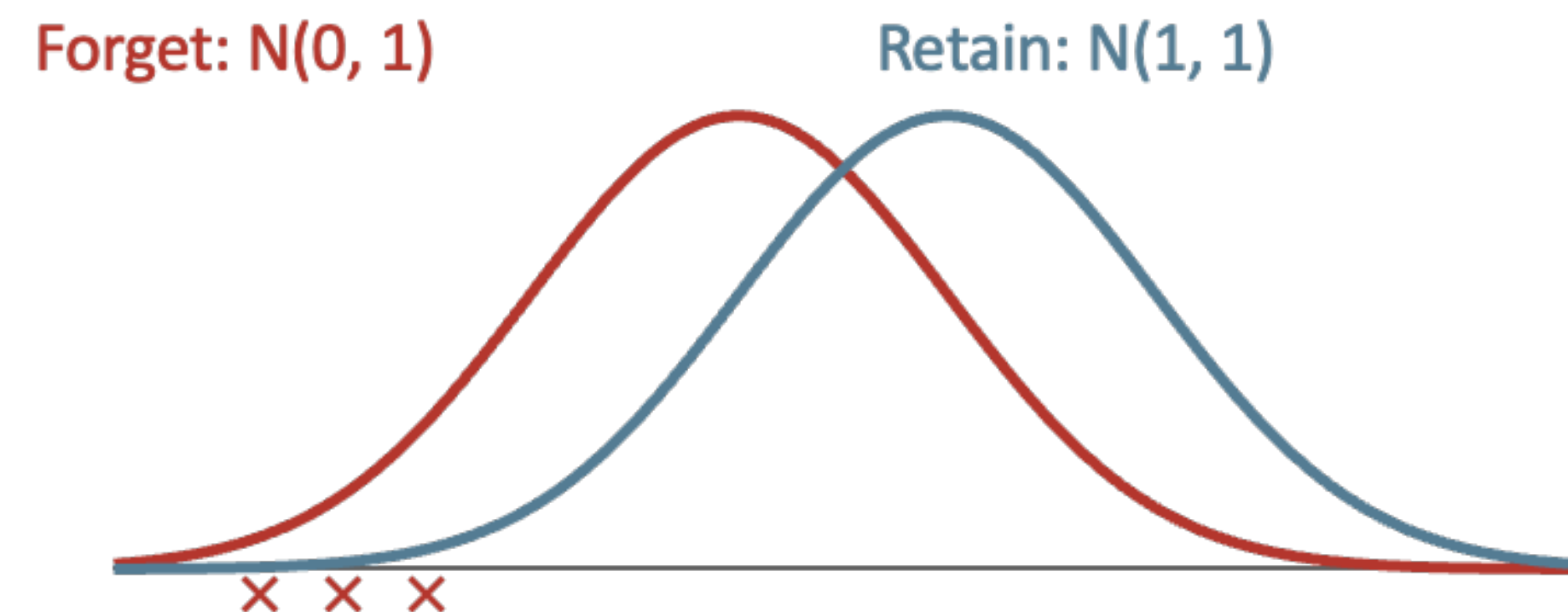
arXiv:2506.06985

Anastasia Koloskova^{* 1} Youssef Allouah^{* 2} Animesh Jha¹ Rachid Guerraoui² Sanmi Koyejo¹

From High-Level Requests to Data Deletion



- How to feed “Forget Harry Potter” to data deletion mechanisms?
- *Idea*: curate examples of the broader “distribution” we want to delete
- *Canonical problem*: two Gaussians



From High-Level Requests to Data Deletion

Published as a conference paper at ICLR 2026

DISTRIBUTIONAL MACHINE UNLEARNING VIA SELECTIVE DATA REMOVAL

arXiv:2507.15112

Youssef Allouah^{1*}, Rachid Guerraoui², Sanmi Koyejo¹

¹Stanford University, USA ²EPFL, Switzerland



Is your algorithm unlearning or untraining?

Eleni Triantafillou¹, Ahmed Imtiaz Humayun¹, Mónica Ribero¹, Alexander Matt Turner¹, Michael C. Mozer¹
and Georgios Kaissis²

¹Google, ²Hasso Plattner Institute

Published as a conference paper at COLM 2025

LLM Unlearning Reveals a Stronger-Than-Expected Coreset Effect in Current Benchmarks

Soumyadeep Pal^{†,*} Changsheng Wang^{†,*} James Diffenderfer[‡]
Bhavya Kailkhura[‡] Sijia Liu^{†,§}

[†]Michigan State University, [‡]Lawrence Livermore National Laboratory, [§]IBM Research

From High-Level Requests to Data Deletion

- A lot more to do!

LEARNING WHAT TO FORGET: DISTRIBUTIONAL UN- LEARNING FOR LLM REPRESENTATION SPACES

Pinaki Mohanty¹, Haoran Tang¹, Maggie Makar², Rajiv Khanna¹

¹Department of Computer Science, College of Science & College of Engineering,
Purdue University, West Lafayette, IN, USA

² Computer Science and Engineering Division,
University of Michigan, Ann Arbor, MI, USA

Statistical Unlearning of Distributions: A Hypothesis Testing Approach

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Evaluation of Forget Set Curation in Language Models

- Natural language makes it tricky to curate the forget data!
- *Request*: “Forget the song Never Gonna Give You Up”



Evaluation of Forget Set Curation in Language Models

- *Measurement*: verbatim text completion
- *Prompt*: complete “We’re no strangers to love”
- *Problem*: we find matching lyrics in at least 147 documents online, e.g., GitHub, StackOverflow, food blogs, etc.



A screenshot of a web browser window displaying a GitHub repository. The browser's address bar shows 'github.com'. The main content area shows a code editor with a dark background and light-colored text. The code is a JSON-like structure with the following content:

```
5   NG= "Never gonna"
6   V1 = (
7       "We're no strangers to love",
8       "You know the rules and so do I",
9       "A full commitment's what I'm thinking of",
10      "You wouldn't get this from any other guy"
11   )
12   Inter1 = (
13       "I just want to tell you how I'm feeling",
14       "Gotta make you understand"
15   )
16   Chorus = (
17       "give you up",
18       "let you down ",
19       "run around and desert you",
20       "make you cry",
```

The GitHub logo is visible in the bottom right corner of the code editor.

CleanSlate

Benchmarking Forget Set Curation

- 4,616 songs and 50 books, with 12,088 questions about their content
- No pair (data curator, data deletion algorithm) works well enough

	Model	Net reduction (%)		QA	Validation Δ (pp)						
		F $\uparrow$	R $\downarrow$	Δ (pp)	GSM8K	BBH	WinoG	CoQA	HEval+	LMBDA	Avg
BM25 pre	GEMMA-3-12B	39.3%	38.4%	-2.3	+3.9	-4.7	+0.2	-6.4	+4.9	-4.9	-1.2
	OLMO-3-32B	43.1%	30.8%	-0.2	+1.5	+0.2	+0.6	+4.6	+0.0	+0.5	+1.2
	LLAMA-3.1-8B	28.9%	51.2%	-0.4	-0.1	+0.1	-0.6	+1.9	+1.2	-1.0	+0.3
	NEMOTRON-9B	14.6%	14.9%	+0.6	-11.1	-4.0	+0.9	+0.6	+0.6	-1.7	-2.4
	QWEN3-8B	1.6%	2.1%	-3.9	-1.4	+1.4	+3.2	-3.5	-6.7	+3.8	-0.5
	OLMO-3-7B	1.0%	2.9%	-0.1	+1.7	-0.1	-0.1	-3.8	-0.6	-0.0	-0.5
	Average	21.4%	23.4%	-1.1	-0.9	-1.2	+0.7	-1.1	-0.1	-0.5	-0.5
BM25 mid	GEMMA-3-12B	46.3%	46.0%	-3.2	+2.0	-8.6	-0.4	-9.2	+2.4	-6.3	-3.4
	OLMO-3-32B	43.0%	35.2%	+0.3	+2.4	+0.3	+1.4	+2.9	+0.6	+0.7	+1.4
	LLAMA-3.1-8B	16.7%	54.0%	-1.1	-0.2	-0.4	-1.5	-2.4	+1.8	-1.4	-0.7
	NEMOTRON-9B	13.0%	14.8%	+0.6	-10.1	-4.1	+0.6	-0.3	+0.6	-2.4	-2.6
	QWEN3-8B	2.3%	2.7%	-5.8	-1.1	+1.4	+3.6	-0.6	-10.4	+3.6	-0.6
	OLMO-3-7B	1.5%	3.7%	-0.0	+1.0	-0.2	+0.0	-1.9	-1.2	-0.3	-0.4
	Average	20.5%	26.1%	-1.5	-1.0	-2.0	+0.6	-1.9	-1.0	-1.0	-1.0
Infingram mid	GEMMA-3-12B	47.8%	44.9%	-2.8	+3.2	-5.3	-0.2	-10.1	+1.8	-6.0	-2.8
	OLMO-3-32B	44.9%	36.0%	+0.4	+2.7	+0.1	+1.7	+3.9	+0.0	+0.9	+1.5
	LLAMA-3.1-8B	19.4%	55.7%	-0.8	+0.3	-0.4	-0.9	+0.9	+0.0	-1.4	-0.2
	NEMOTRON-9B	12.7%	14.5%	+0.5	-0.1	-2.3	+0.2	+0.3	+3.7	-2.3	-0.1
	QWEN3-8B	2.8%	2.8%	-4.6	-1.0	+1.5	+3.9	-3.1	-9.8	+3.6	-0.8
	OLMO-3-7B	1.6%	3.2%	-0.1	+0.9	-0.3	+0.0	-0.8	-0.6	-0.1	-0.2
	Average	21.6%	26.2%	-1.2	+1.0	-1.1	+0.8	-1.5	-0.8	-0.9	-0.4
EA	GEMMA-3-12B	96.2%	85.4%	-6.1	+3.6	-29.4	+0.0	-6.3	+1.2	-13.8	-7.4
	OLMO-3-32B	100.0%	97.0%	-2.7	+3.2	-3.3	-4.9	+3.9	-1.2	-3.0	-0.9
	LLAMA-3.1-8B	100.0%	100.0%	-5.7	-11.8	-37.6	-3.1	-3.9	+3.7	-10.1	-10.4
	NEMOTRON-9B	100.0%	80.3%	-1.2	-23.1	-22.8	+0.2	-5.5	-1.2	-5.2	-9.6
	QWEN3-8B	100.0%	82.7%	-21.9	-3.0	-2.0	+0.3	-7.9	-4.9	-6.0	-3.9
	OLMO-3-7B	100.0%	72.2%	-1.0	+2.8	-0.1	-2.0	-5.4	+1.8	-2.3	-0.9
	Average	99.4%	86.3%	-6.5	-4.7	-15.9	-1.6	-4.2	-0.1	-6.7	-5.5

CleanSlate

Benchmarking Forget Set Curation

- *Open Problem:* build a forget set curator that works well on our benchmark

What to Forget in Unlearning? Forget Set Curation for Language Models

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